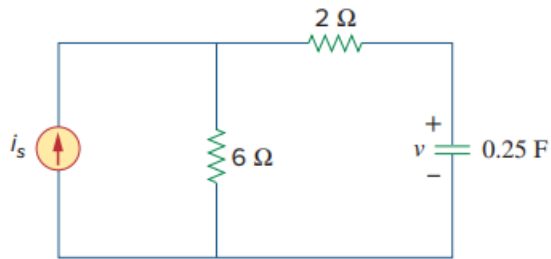


Problem

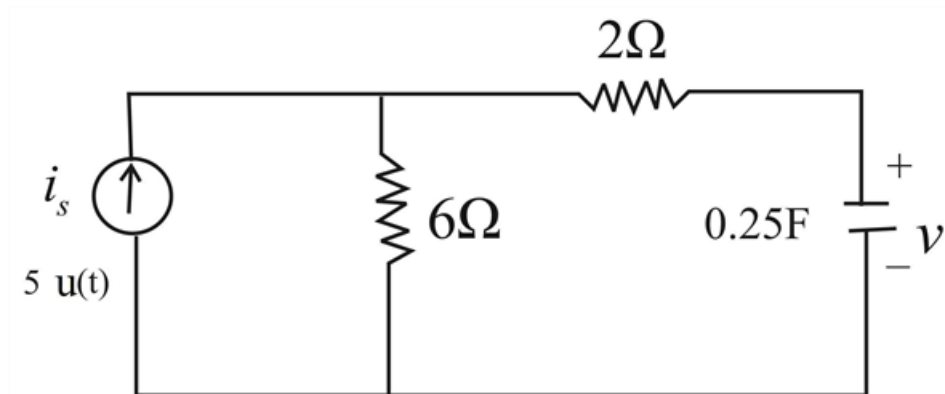
For the circuit in Fig. 7.113, $i_s(t) = 5u(t)$. Find $v(t)$.

Fig. 7.113:



Step-by-step solution

Step 1 of 5

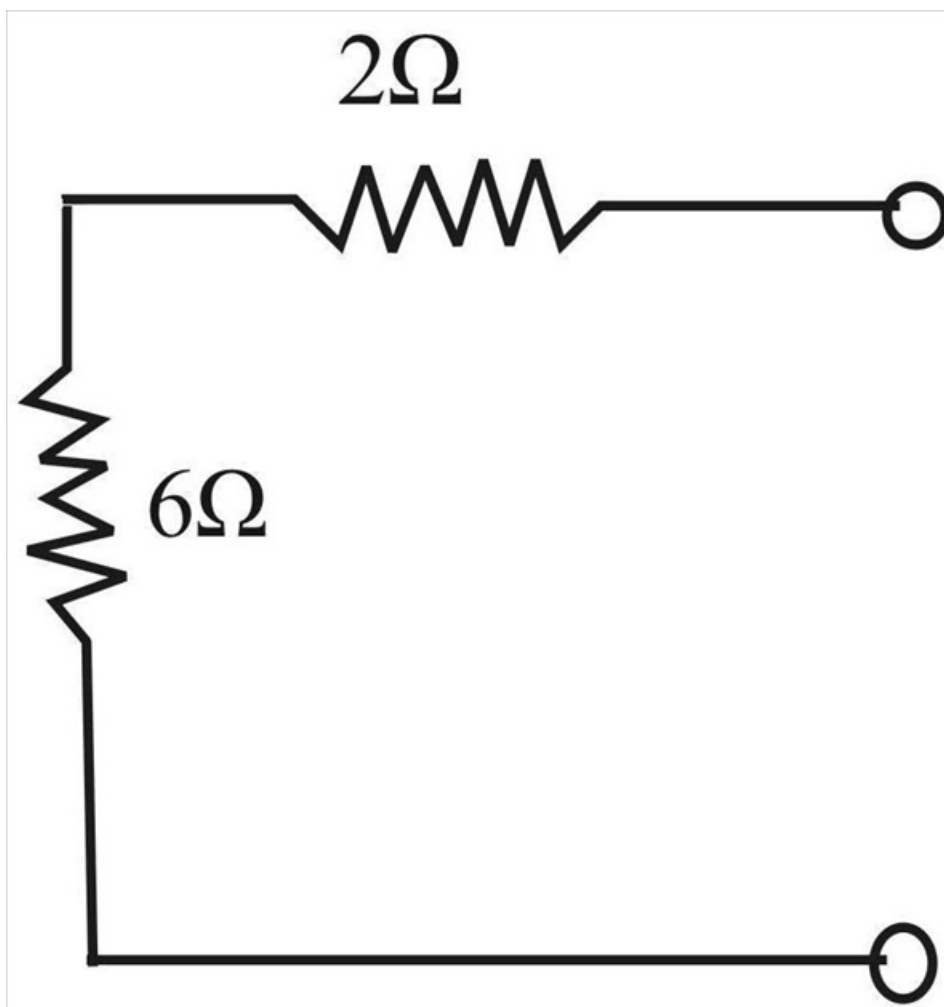


Step 2 of 5

We have to find $V(t)$

$$\therefore V_C(t) = A - (A - B) e^{\frac{-t}{\tau}}$$

Step 3 of 5



Step 4 of 5

Where,

$$A = V_C(\infty)$$

$$= 6 \times 5$$

$$= 30 \text{ V}$$

$$B = V_C(0^+)$$

$$= V_C(0^-)$$

$$= 0$$

Step 5 of 5

$$\tau = R_{eq} \cdot C$$

$$= (6 + 2)(0.25)$$

$$= 8 \times 0.25$$

$$= 2 \text{ s}$$

So, $v_C(t) = 30 - (30 - 0) e^{\frac{-t}{2}}$

$$\boxed{v_C(t) = 30(1 - e^{-0.5t}) \text{ V}}$$